

Differential Pairs

Conductor Width (W)

0.16 mm

Conductor Spacing (S)

0.2 mm

Conductor Height (H)

0.0994 mm

Applied Voltage

1 Volts

Coupled Length

2.54 mm

Signal Risetime

1 ns

Target Zdiff

90 Ohms

+/- Tolerance = 10%



Target Zdiff Plus

101.250 Ohms

Target Zdiff Minus

78.750 Ohms

Formula Restrictions:

$0.1 < W/H < 3.0$

$0.1 < S/H < 3.0$

Differential Protocol

USB 3.X

Send to Via Calculator

Zdifferential

89.235 Ohms

Zo

47.953 Ohms

Zodd

44.618 Ohms

Zeven

51.538 Ohms

? Help

Kb (Term)

0.0360

dB

-28.866

NEXT Voltage

0.0011 V

Kb' (Unterm)

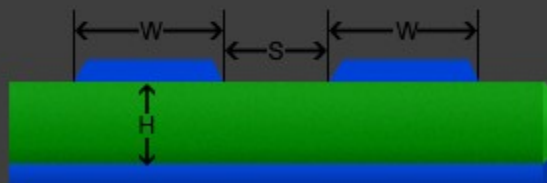
0.0720

dB

-22.857

NEXT Voltage

0.0021 V



Options

Base Copper Weight

- ☐ 9um
- ☐ 18um
- ☒ 35um
- ☐ 53um
- ☐ 70um
- ☐ 88um
- ☐ 106um
- ☐ 142um
- ☐ 178um

Plating Thickness

- ☒ Bare PCB
- ☐ 18um
- ☐ 35um
- ☐ 53um
- ☐ 70um
- ☐ 88um
- ☐ 106um

Differential Layer

- ☒ Edge Cpld Ext
- ☐ Edge Cpld Int Sym
- ☐ Edge Cpld Int Asym
- ☐ Edge Cpld Embed
- ☐ Broad Cpld Shld
- ☐ Broad Cpld NShld

Information

Total Copper Thickness
35 um

W/H = 1.610

S/H = 2.012

Lsat = 85.35 mm

Units

- ☐ Imperial
- ☒ Metric

Substrate Options

FR-4 STD

Er

4.1

Tg (°C)

130

Temp Rise (°C)

20

Temp in (°F) = 36.0

Ambient Temp (°C)

22

Temp in (°F) = 71.6

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Solve!



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